BANK CUSTOMER CHURN ANALYSIS

SQL Query Report

Portfolio Project

Analyzed 10,000 bank customer records to identify key factors driving customer churn across geography, demographics, engagement, product usage, and account balance.

Original SQL code & result screenshots preserved

Bank Churn Analysis



Objectives

This report presents insights from a Bank Customer Churn Analysis conducted on 10,000 customer records. SQL (SQLite) was used for data preparation and analysis, while Tableau Public was used to develop an interactive dashboard. The objective was to identify the major drivers of churn and provide business recommendations to improve.

Query1:

```
--1. Overall churned ?
select count(*) as total_customers , sum(exited) as total_churned,
 sum(exited)*100 /count(*), 2) as overall_churned from Bank_churn;
 -- we used sum(exited) coz exited are in 0,1 and know 0= stayed and 1= churned(left)
 --so by doing this we can find total churned just by doing this
```

- By this query we came to know that in our 10000 customers there are 20% churned customers.

total_customers	total_churned	overall_churned
10000	2037	20.0

- This will help us to focus on that churned customers , why they are leaving rather than on 80% or 100%.

Query 2:

```sql
--2. Churn rate by country ?
select Geography, count(*) as total_customer , sum(Exited) as total_churned ,
round(sum(exited)*100/count(*),2) as country_churned
 from Bank_churn group by Geography order by country_churned DESC;
```

- By this query we came to know that in our bank we have 32% churned customers from Germany, 16% from Spain and 16% from France.

Geography	total_customer	total_churned	country_churned
Germany	2509	814	32.0
Spain	2477	413	16.0
France	5014	810	16.0

- As we can see that customer of Spain and Germany is almost same but the churn rate of Germany is double than Spain ,means Spain is performing better.Germany should be prioritized for investigation.

Query 3:

```sql
--3. Churn rate by Gender?
select  Gender , count(*) as total_customer , sum(exited) as total_churned ,
sum(exited)*100/count(*) as churn_rate from Bank_churn group by gender;
```

- From this query we came to know that in our bank, there is more female churn rate than male churn rate.
- Female customers have a higher observed churn rate than male customers in this dataset, surely further investigation into potential differences in customer experience and product needs.

Gender	total_customer	total_churned	churn_rate
Female	4543	1139	25
Male	5457	898	16

Query 4:

- The objective of this query is to find out the churn rate of customers in different age group.

```
select case
   when Age < 20 then 'Under_age'
   when Age BETWEEN 20 and 30 then '20-30'
   when Age BETWEEN 30 and 35 then '30-35'
   when Age BETWEEN 35 and 40 then '35-40'
   when Age BETWEEN 40 and 45 then '40-45'
   when Age BETWEEN 45 and 50 then '45-50'
   when Age BETWEEN 50 and 60 then '50-60'
   ELSE 'Above 60'
   END AS AGE_GROUP, count(*) as total_customer , sum(exited) as total_churn
sum(exited)*100/count(*) as churn_rate from Bank_churn group by AGE_GROUP
order by churn_rate desc ;
```

- Older customer segments show substantially higher churn rates. Further investigation is required to determine whether product suitability, service experience or other factors contribute to this pattern.

AGE_GROUP	total_customer	total_churned	churn_rate
50-60	797	448	56
45-50	850	394	46
40-45	1470	394	26
Above 60	464	115	24
35-40	2266	339	14
30-35	2185	199	9
20-30	1919	145	7
Under_age	49	3	6

Query 5:

```sql
--5. Do active members churn less?
--in isactivemembers:
--1 → customer actively uses the bank (logs in, makes transactions, uses services)
-- 0 → customer has an account but rarely uses it (inactive)
select IsActiveMember , count(*) as total_members , sum(Exited) as total_churn ,
sum(Exited) *100/ count(*) as churn_rate from Bank_churn group by IsActiveMember
order by churn_rate desc;
```

- From the above query, we can see that there are active and inactive members of the bank.
- Although active members have a lower churn rate 14% than inactive members 26%, the fact that 14% of actively engaged customers still churns point of further investigation. The bank should analyze customer experience, product satisfaction, service quality, and competitive factors to understand why engaged customers are leaving.

IsActiveMember	total_members	total_churn	churn_rate
0	4849	1302	26
1	5151	735	14

Query 6:

```sql
--6.churn by number of products?
select NumOfProducts, count(*) as total_members , sum(Exited) as total_churn ,
round(sum(Exited) *100/ count(*) , 2)as churn_rate
 from Bank_churn group by NumOfProducts
order by churn_rate desc;
```

- The objective of this query is to find churn rate of the people who is using many services/product of our bank.
- The 4-product segment shows a 100% churn rate, but the segment contains only 60 customers. This should further investigate before coming up with conclusion.

NumOfProducts	total_members	total_churn	churn_rate
4	60	60	100.0
3	266	220	82.0
1	5084	1409	27.0
2	4590	348	7.0

Query 7:

```sql
--7.  High Balance Customers Who Churned?
select case
when Balance = 0 then 'zero balance'
 when Balance < 50000 then 'low balance(under 50k)'
 when Balance BETWEEN 50000 and 100000 then 'Mid Balance(50k-100k)'
 ELSE 'High balance(<100k)'
 END as Type_of_balance,
 count(*) as total , sum(Exited)as churned ,
 Round(sum(exited)*100 / count(*) , 2) as churned_rate
 from Bank_churn
 group by Type_of_balance
 order by Type_of_balance ;
```

- The objectives of this query was to see churn rate of customers according to there bank balance.

• Low-balance customers have the highest observed churn rate at 34%, followed by high-balance customers at 25%.Further analysis is required to understand whether

Type_of_balance	total	churned	churned_rate
High balance(<100k)	4799	1211	25.0
Mid Balance(50k-100k)	1509	300	19.0
low balance(under 50k)	75	26	34.0
zero balance	3617	500	13.0

income, product usage, customer tenure, or other factors contribute to the difference.